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Vendor selection in the wake of data breaches: A longitudinal study

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Abstract

With the increasing digitization and networking of medical data and personal health information, information security has become a critical factor in vendor selection. However, limited understanding exists regarding how information security influences vendor selection. Drawing from the attention-based view (ABV), this study examines the potential impact of data breaches on hospitals' selection of electronic medical record system (EMRS) vendors. To test our hypotheses, we compile a unique dataset spanning 12 years of observations from US hospitals. Utilizing a coarsened exact matching (CEM) technique combined with a difference-in-differences (DiD) approach, our study shows that hospitals tend to replace their EMRS vendors after experiencing data breaches. Moreover, breached hospitals tend to prioritize information security in such a vendor replacement process by switching to star vendors and migrating towards a single-sourcing configuration. Further post-hoc analyses reveal that these impacts of data breaches are mitigated as the relationship between breached hospitals and vendors matures or when hospitals belong to large healthcare systems. Additionally, we find that the effects of data breaches are contingent on the scale of the breach and are short-term in nature. This research underscores the significance of information security as a crucial consideration in vendor selection for both academia and practitioners.

KEYWORDS

attention-based view, data breach, electronic medical record systems, information security, information technology outsourcing, vendor selection

Highlights

- We find that hospitals tend to change their electronic medical record system (EMRS) vendors following data breaches. During this replacement, they are more inclined to select star vendors and migrate towards a single-sourcing configuration.
- We also find that the impacts of data breaches can be mitigated as the relationship between breached hospitals and vendors matures, or when hospitals are integrated into larger healthcare systems.
- Additionally, we find that the effects of data breaches are dependent on the scale of the breach and are typically short-term in nature.

1 | INTRODUCTION

Vendor selection plays a pivotal role in the successful operation and growth of organizations (Stuart, 1993; Wetzstein et al., 2016). It encompasses the careful assessment and evaluation of potential vendors to ensure that they align with the organizations' requirements and objectives (Sonmez, 2006; Verma & Pullman, 1998). By selecting vendors with the appropriate expertise, resources, and capabilities, organizations can leverage the vendors' strengths to meet customer expectations and drive business growth (Kannan & Tan, 2002; Vonderembse & Tracey, 1999). Additionally, a well-executed vendor selection process facilitates the establishment of long-term partnerships and stable supply relationships, ensuring a consistent and reliable flow of goods or services (Ellram, 1990).

The existing literature has highlighted numerous key principles that organizations may consider when selecting third-party vendors, such as cost, efficiency, collaboration history, and professional capabilities (Dohmen & van Raaij, 2019; Gupta & Barua, 2018; Kale & Zollo, 2005; Schiele, 2006; Wadhwa & Ravindran, 2007). However, in the current digital era, information security concerns have emerged as a new sticking point for clients in establishing relationships with their third-party vendors (Dhillon et al., 2017; Khalfan, 2004). The reason is that outsourcing often involves delegating information technology (IT) functions and granting long-term privileged access to third-party vendors, allowing the latter to operate the outsourced IT functions remotely. This situation will lead to the fragmentation of security measures and poses challenges for client organizations in closely monitoring off-site operations, as they typically would with on-site teams. Despite this, there is a lack of understanding regarding how information security influences vendor selection. To address this research gap, this study aims to undertake a pioneering investigation into the role of information security in vendor selection.

We specifically observe a phenomenon related to the attitude towards information security in vendor selection processes within organizations. Despite significant concerns about information security when collaborating with third parties, organizations often tend to overlook these concerns during the vendor selection process. A survey conducted by the Ponemon Institute (2019) reveals that many organizations have not adequately embraced risk assessments to accurately measure or manage the information security threats posed by vendors. Moreover, in this survey, approximately half of the respondents indicate that their organizations do not even require third parties to demonstrate compliance with privacy and security regulations. Similarly, another survey conducted by

the Ponemon Institute (2018) shows that numerous organizations have shared sensitive data with vendors without fully acknowledging or assessing their security policies. However, this trend is likely to change following an organization's experience of an information security failure (i.e., data breach). In the aftermath of a data breach, information security concerns will become prominent and potentially reshape the decision-making process of decision-makers (Snedaker, 2013; Solove & Citron, 2017). This observation leads to the following questions: *Will data breaches drive organizations to change their vendors? If so, how will they go about making these changes?*

We particularly concentrate on a single context in answering the aforementioned research question: the outsourcing of electronic medical record systems (EMRS) in US hospitals, which involves transferring EMRS operations directly to third parties. Focusing on this EMRS outsourcing context offers several advantages for our study. First, it helps control for additional contextual factors that could potentially impact organizations' vendor selection processes (Angst et al., 2017). Second, in the current EMRS supply market, hundreds of EMRS vendors competing with each other, providing organizations with a wide range of choices to tailor their EMRS vendor selection strategies to their specific needs and preferences (IBISWorld, 2014). Third, there is a global trend where medical organizations are transitioning from paper-based medical records to outsourcing their EMRS (Enaizan et al., 2020; Poursaghar et al., 2008). Despite the significant increase in EMRS usage in practice, the academic understanding of EMRS vendor selection remains ambiguous. Therefore, our focus on the background of EMRS outsourcing holds important academic and practical implications.

We theoretically analyze the impact of data breaches on the EMRS vendor selection strategy through the lens of the attention-based view (ABV) (Barnett, 2008; Ocasio, 1997). The ABV perspective is selected due to its widespread use in examining how organizations divert their attentions at making decisions in different contexts (Ocasio et al., 2018; Panagiotopoulos & Bowen, 2020; Rhee & Leonardi, 2018). The fundamental principle of ABV is that decision-makers in organizations have limited attention capacity, leading them to allocate attention based on problem salience. Problem salience varies in different contexts, driving decision-makers to shift attention and make adjustments. In our context, a data breach can serve as a stimulus that creates cognitive dissonance for decision-makers and reminds hospitals of potential security shortcomings, altering the salience of information security principle. As security principles become more prominent, decision-makers tend to focus on identifying

existing security vulnerabilities and seeking solutions (Benaroch & Chernobai, 2017; Talbot & Jakeman, 2011). Therefore, we predict that hospitals experiencing data breaches are more likely to uncover deficiencies in their current EMRS vendors and switch to alternative vendors, compared to normal operating conditions without data breaches. Furthermore, during the vendor replacement process, we predict that hospitals will favor patterns that are more beneficial for information security, such as choosing highly recognized vendors (e.g., star vendors) or migrating towards a single-sourcing configuration to minimize the potential scope of attack.

We empirically test our predictions using a longitudinal dataset covering 12 years (2006–2017) from US hospitals. To identify the causal impact of data breaches on vendor selection behavior, we employ a rigorous identification strategy, namely, a coarsened exact matching (CEM)-combined-difference-in-differences (DiD) approach. In this approach, we treat hospitals-years that experience data breaches as the treatment group, while hospitals-years without data breaches serve as the control group. We then compare changes in vendor selection behaviors between the pre- and post-breach periods in the treatment and control groups. We also utilize the CEM technique to enhance comparability between the two groups. Our analysis results provide strong evidence that supports the hypotheses put forth in this study. Specifically, we find that hospitals are more likely to replace their EMRS vendors after experiencing a data breach. During the vendor replacement process, they show a significant preference for switching to star vendors and migrating towards a single-sourcing configuration. Additionally, in our post-hoc analysis, we explore contingency factors related to the relationships between data breaches and vendor selection behaviors. We find that these relationships are further weakened as breached hospitals' relationships with vendors mature or when they are part of large hospital systems. Moreover, we observe that the impact of data breaches on vendor selection behaviors frequently depends on the scale of the breach and exhibits short-term effects.

Our study holds significant implications. First, it makes a significant contribution to the existing literature on vendor selection. While the previous vendor selection research has focused on criteria such as cost, vendor efficiency, cooperation history, and professional competence (see a literature review in Ho et al., 2010), the importance of information security has been largely overlooked. This study fills the gap in the literature by identifying information security as a vital criterion in shaping organizations' vendor selection behaviors. Second, our study contributes to the exploration of information security within the field of operations management (OM). While information

security is widely recognized as a critical concern in supply chain management (Ghadge et al., 2020; Pandey et al., 2020), there is a dearth of empirical research on this topic within the OM field. Our study fills this research gap and advances the field by pioneeringly providing empirical evidence on the impact of information security failures on supply chain collaboration. Third, this study contributes to the ABV literature by examining how organizations' negative performance, specifically data breaches, serve as stimuli to divert attention and drive strategic changes in the supply chain context. Lastly, this study contributes to the OM literature on multi-sourcing by identifying information security as a crucial determinant of organizations' multi-sourcing decisions, filling a research gap in the limited understanding of the factors that influence multi-sourcing.

2 | RESEARCH BACKGROUND AND LITERATURE REVIEW

2.1 | Research background

We concentrate on the selection of EMRS vendors by hospitals. In the current landscape, paper-based medical records are quickly becoming outdated due to their inherent disadvantages, including inefficiency, vulnerability to loss, high handling costs, interpretation difficulties, and susceptibility to damage. These drawbacks have prompted a global shift towards EMRS, which are information systems designed to manage patients' digital medical records (Boonstra & Broekhuis, 2010). Extensive research has demonstrated that organizations' adoption of EMRS leads to cost reduction, decreased medical errors and adverse drug events, as well as increased adherence to best practices (Boonstra & Broekhuis, 2010; Hillestad et al., 2005).

A notable trend in hospitals is the prevalent practice of outsourcing their EMRS to third-party providers. This allows organizations to decrease their operational workload and enables physicians to utilize their time more efficiently, rather than spending extensive periods in front of computers. As a result of this outsourcing trend, numerous EMRS providers now operate in the EMR management space (Angst et al., 2017). Within this context, hospitals have the opportunity to select appropriate EMRS vendors by assessing their compatibility with the organization's core principles and requirements.

However, outsourcing any aspect of a business, including EMRS, presents inherent risks for hospitals. One of the primary concerns in an outsourcing environment is information security (Dhillon et al., 2017; Khalfan, 2004). Given the sensitive nature of EMRS and

the regulatory requirements outlined in the Health Insurance Portability and Accountability Act (HIPAA), it is crucial for hospitals to ensure the secure storage, backup, transmission, and disposal of their electronic medical records. However, when outsourcing EMRS, hospitals relinquish control of EMRS controls to an off-site company, potentially resulting in a loss of granular control over related operations and functions. Moreover, due to the information asymmetry in supply chain collaborations, hospitals often face challenges in ensuring that their partnering vendors effectively adhere to security regulations and protocols (Bhatti et al., 2017). Despite these challenges, in the context of EMRS management, hospitals are required to grant extensive remote access and management privileges to their vendors as part of the delegated agency relationship. However, such a grant of access means that any mistake or violation on the part of the third-party vendors could directly compromise the clients' sensitive information.

Against the aforementioned backdrop, hospitals need to consider information security when collaborating with EMRS vendors. Nevertheless, practical evidence indicates that many organizations often neglect to adequately assess information security issues during the vendor selection process. For example, a survey conducted by Soha Systems (2016) reveals that only 2% of IT professionals consider secure data access by vendors as a top priority, despite the increasing number of security threats associated with third-party data access. The PWC Global State of Information Security report (2018) indicates that only two out of five hospitals require their third-party contractors to comply with their privacy policies. Ed Gaudet, CEO and Founder of Censinet, summarizes this phenomenon by stating that "the overall process for managing risk assessments is severely broken in health-care."¹ In light of this context, our study investigates whether information security failures likely prompt hospitals to make adjustments in their selection of EMRS vendors to address potential oversights regarding information security.

2.2 | Literature review

There are two streams of OM literature that directly relate to our study: (1) vendor selection and (2) digital information security. Vendor selection is a central topic in OM research, with most studies relying on analytical modeling techniques, ranging from simple matrix approaches to more advanced mathematical programming methods (Chai et al., 2013; Chai & Ngai, 2020; Petroni & Braglia, 2000; Weber et al., 1991). However, empirical research in this domain is relatively limited but

highly relevant to our study. Thus, we conduct a review of empirical studies that examine vendor selection behaviors in the OM field. These studies focus on exploring novel factors that influence vendor selection behaviors. They can be broadly categorized into two strands based on the types of vendor selection behaviors that they examine. The first strand explores factors influencing preferences in vendor selection, such as procedural rationality (Riedl et al., 2013), economies of scale (Badorf et al., 2019), cycle time (Sharland et al., 2003), social media volume and valence (Banerjee et al., 2020), social sustainability (Thomas et al., 2021), and supplier ties (Yan et al., 2020). The second strand investigates factors driving customers to switch from existing vendors, such as trustworthiness (Ta et al., 2018), responsibility attribution of supplier-induced disruptions (Wang et al., 2022), clients' post-disruption emotion (Polyviou et al., 2018) and relationship quality and switching costs (Whitten & Leidner, 2006). In Appendix A, we provide a comprehensive list of relevant literature and summarize their insights. Our study aims to extend this literature by introducing the consideration of information security factors.

This study is also relevant to the existing literature on digital information security in the field of OM. With the increasing prevalence of big data, information security has become a critical requirement in OM, attracting growing interest from OM scholars. The research in the OM field on digital information security focuses on both upstream (antecedents) and downstream (consequences) aspects of organizations' digital information security. Regarding the *upstream* aspects, Massimino et al. (2017) examine the influence of location-related factors on firms' confidentiality performance and find that industrial agglomeration is associated with improved confidentiality performance at the developer level. Lan et al. (2020) investigate the impact of firms' network positions on their product and confidentiality performance. They discover that network centrality is positively related to product performance but negatively related to confidentiality performance, and additionally, firms with more structural holes demonstrate better performance in terms of both product and confidentiality. Regarding the *downstream* aspects, Modi et al. (2015) analyze the market reaction to service information failures of providers using an event study methodology. They find that service information failures caused by providers result in greater financial losses compared to failures attributed to client organizations. Ried et al. (2021) examine the spillover effects of organizations' information leakage, showing that such leakage reduces observers' willingness to share information with the offenders, especially after opportunistic leakages. However, the literature on digital

information security in the OM field remains limited in scope. Massimino et al. (2018) highlight this research gap, stating that “digital confidentiality has received significantly less research attention from operations and supply chain management (OSCM) scholars compared to cost, quality, flexibility, delivery, and innovation” (p. 1492). To address this research gap, our study pioneers the exploration of how organizations respond to security failures in the context of vendor selection.

3 | THEORETICAL BACKGROUND AND HYPOTHESES

3.1 | Attention-based view (ABV)

This study provides a theoretical explanation for the impact of data breaches, utilizing the ABV theoretical framework. This framework offers valuable insights into how organizational decision-makers allocate their attention to address a range of principles and issues they face (Ocasio, 1997; Sullivan, 2010).

According to the ABV framework, decision-makers in organizations have limited attentional resources, and different issues compete for their attention (Ocasio, 1997). When allocating their attention, decision-makers tend to prioritize simpler, less complex, and routinized issues that offer immediate feedback (Sullivan, 2010). However, the allocation of attention by decision-makers is context-dependent and not determined by knowledge alone. Therefore, what problem or principle that they “focus on depends on the specific situation” (Ocasio, 1997: 187). Furthermore, the ABV framework highlights the notion that “what decision-makers do depends on what issues and answers they focus their attention on” (Ocasio, 1997, p. 187). In other words, decision-makers tend to develop strategies and undertake subsequent actions based on how they allocate their attention.

Our theorizing integrates such an ABV perspective and considers data breaches as a stimulus to change decision-makers' attention allocation, consequently shaping organizational move, as discussed in the following sections.

3.2 | Whether to change EMRS vendors in the wake of data breaches?

To begin, we examine the impact of data breaches on hospitals' decisions to change EMRS vendors from an ABV perspective (Ocasio, 1997; Sullivan, 2010). Under normal operating conditions (i.e., without data breaches), decision-makers in hospitals tend to prioritize aspects

directly associated with generating high returns for shareholders. These aspects are routine, direct, and closely aligned with their primary responsibilities, thus receiving higher priority in attention allocation according to ABV (Sullivan, 2010). While security principles may also be among the issues that decision-makers focus on, the uncertain returns associated with security investments suggest that these principles may receive lower priority in attention allocation (Sullivan, 2010). This tendency is also extensively supported by anecdotal evidence. For example, according to Bomgar's (2016) survey, numerous businesses consider costs more important than security, with 64% of respondents explicitly stating that their organizations prioritize cost over security when selecting third-party vendors.

However, the occurrence of data breaches deviates decision-makers' expectations from the normally held ones and forces them to reallocate their attention in view of the recent expectancy violation (Benaroch & Chernobai, 2017). Organizations' failures in terms of information security may break decision-makers' false sense that their organizations are highly secure, reveal potential weaknesses and vulnerabilities in their IT systems, remind organizations of the significance of information security, and heighten decision-makers' concerns in this regard (Nikkhah & Grover, 2022). As a result, security principles within organizations are brought to the forefront as a key element in capturing decision-makers' attention.

From an ABV perspective (Ocasio, 1997; Sullivan, 2010), the increased focus on information security can motivate decision-makers to proactively identify any shortcomings and make the necessary improvements to enhance performance in this specific area. Considering that EMRS vendors bear direct responsibility for managing sensitive hospital information, their security performance becomes a pivotal consideration. Consequently, decision-makers may conduct a comprehensive reassessment of their EMRS vendors' security capabilities and qualifications. Compared to normal operating conditions without data breaches, decision-makers are more likely to recognize security deficiencies in their current EMRS vendors and actively seek alternatives with superior information security capabilities. Additionally, as previously mentioned, hospitals' limited attention often results in a relative neglect of information security principles and a sense of complacency about their security performance in normal conditions. However, security failures serve as stark reminders that hospitals can no longer disregard information security, compelling them to pursue transformative changes (e.g., replacing EMRS vendors) to improve their security performance. This expectation of EMRS vendor replacement in the wake of data breach

has empirical support in the research documenting a positive relationship between adverse organizational events (e.g., product recalls, medical errors, IT failures, and major disasters) and organizational change (e.g., business innovation, alterations to governance, and process improvement) (Benaroch & Chernobai, 2017; Crossan & Berdrow, 2003; Dahlin et al., 2018; Sosna et al., 2010). Based on these insights, we propose our first hypothesis as follows:

Hypothesis 1. Data breaches are positively associated with EMRS vendor replacement.

3.3 | How to change EMRS vendors in the wake of data breaches?

We also aim to examine the specific strategies that breached hospitals may employ when changing their EMRS vendors. While a sourcing approach involves a range of strategies, such as location decisions (offshore vs. domestic), make-or-buy decisions, and quality of purchases, this study specifically focuses on two outsourcing strategies in terms of (1) the selection of vendors and (2) the number of vendors to be used.

3.3.1 | Which vendors to select?

We may question which EMRS candidates will be selected as new vendors to replace the old ones in the wake of data breaches. To find answers, we turn to the ABV perspective (Ocasio, 1997; Sullivan, 2010). As mentioned earlier, data breaches can act as reminders to managers regarding the importance of information security, prompting them to pay specific attention to related issues. From the ABV perspective, breached hospitals are likely to take actions aligned with this attentional direction, which in this case, involves seeking solutions to bolster their information security and prevent future security failures. Therefore, as decision-makers select their new third-party EMRS vendors in the wake of data breaches, the intense focus on security may drive decision-makers to prefer candidates who perform well in information security, such as those with a high information security capability or the potential to provide significant security benefits to clients. However, this situation presents a problem: How can decision-makers ascertain the security performance of candidate EMRS vendors? This issue is problematic because candidates' security standards are not outwardly visible. Therefore, decision-makers typically search for cues that reveal something about them (Gomulya & Mishina, 2017).

One particular means by which decision-makers assume or draw inferences on a vendor's good information security performance is by evaluating whether or not the vendor has received prestigious awards from recognized third-party organizations (i.e., are "star" vendors). This inference is based on several reasons. First, the literature has suggested that when an identified third-party organization disseminates positive evaluations of products or services, clients' trust and perceptions of the endorsed products or services increase (Allen, 1984; Bapna et al., 2016). Particularly, an important category to rank EMRS vendors frequently involves security- and privacy-related aspects. As a result, client hospitals tend to trust that star vendors possess notable capabilities in safeguarding information security. Second, hospitals tend to believe that star EMRS vendors have strict contractual obligations to their clients in order to maintain their reputation in the marketplace (Hauser, 2017). Thus, these vendors tend to have well-documented and enforced control procedures (e.g., backup data plans), thereby indicating a relatively high level of information security when engaging in collaborations with them. Third, the recognition obtained through prestigious awards enables vendors to expand their client base (Fernandez-Gamez et al., 2016), resulting in the accumulation of substantial security-related experience and an enhanced ability to control security. Additionally, due to their popularity, star vendors have the capability to share valuable insights with clients confidentially on how other clients have successfully addressed similar security issues. This exchange of knowledge, facilitated by star vendors' high recognition, further amplifies their security benefits for client hospitals.

Overall, the preceding arguments suggest that when decision-makers select new EMRS vendors to replace the old ones after data breaches, the salient information security principles may drive them to highly appreciate star EMRS vendors, the role of whom can be used as an indicator of being capable and beneficial in information security. Thus, we hypothesize as follows:

Hypothesis 2. Data breaches are positively associated with star EMRS vendor switch.

3.3.2 | How many vendors to use?

An additional crucial strategy encountered in EMRS outsourcing is the number of vendors to be used (i.e., single-sourcing vs. multi-sourcing), considering that modern EMRS consists of multiple modules designed for individual specialties (Angst et al., 2017). Single-sourcing involves entrusting all services to a single vendor, while

multi-sourcing (or “best-of-breed”) is an arrangement where services are outsourced to multiple vendors and integrated (Handley et al., 2022). Over the past decades, organizations have gradually shifted away from traditional single-sourcing towards multi-sourcing to leverage diverse external capabilities and benefits, such as cost reduction and avoidance of vendor lock-in (Davies et al., 2013; Levina & Su, 2008).

However, the use of multiple EMRS vendors poses operational challenges, particularly in terms of information security, for the following reasons. First, a multi-sourcing (vs. single-sourcing) arrangement involves changing the status of more outsiders to privileged insiders, each of whom could potentially become a source of information leakage. Consequently, a multi-sourcing arrangement introduces more security vulnerabilities compared to a single-sourcing arrangement. Second, compared with EMRS single-sourcing, the utilization of EMRS multi-sourcing places a greater burden on hospitals in terms of security monitoring and coordination, leading to increased security challenges. This is because hospitals are tasked with the responsibility of consistently monitoring their security levels with each vendor, and as the number of vendors involved grows, the situation becomes more challenging and intricate (Plugge, 2018). Third, the best-of-breed (i.e., multi-sourcing) arrangement has security disadvantages as it introduces complexity to users' security operations. This complexity arises from the fact that multi-sourcing integrated systems consist of multiple independent and diverse applications that are customized to meet the specific requirements of each end user (Hermann, 2010). Consequently, the use of these integrated systems significantly complicates security operations. For instance, when establishing connections with multiple EMRS vendors, employees often encounter challenges in standardizing user logins and passwords across various applications, thereby creating additional security challenges in operations.

The preceding discussion indicates that the use of a multi-sourcing EMRS arrangement poses greater risks to information security compared to single-sourcing. Therefore, as decision-makers' attention flowing into identifying solutions for improving information security in the wake of data breach, they are more likely to move away from multi-sourcing towards single-sourcing. Accordingly, we formulate the following hypothesis:

Hypothesis 3. Data breaches are positively associated with EMRS single-sourcing migration.

Figure 1 illustrates the conceptual framework developed for this research.

4 | SAMPLE SELECTION AND METHODOLOGY

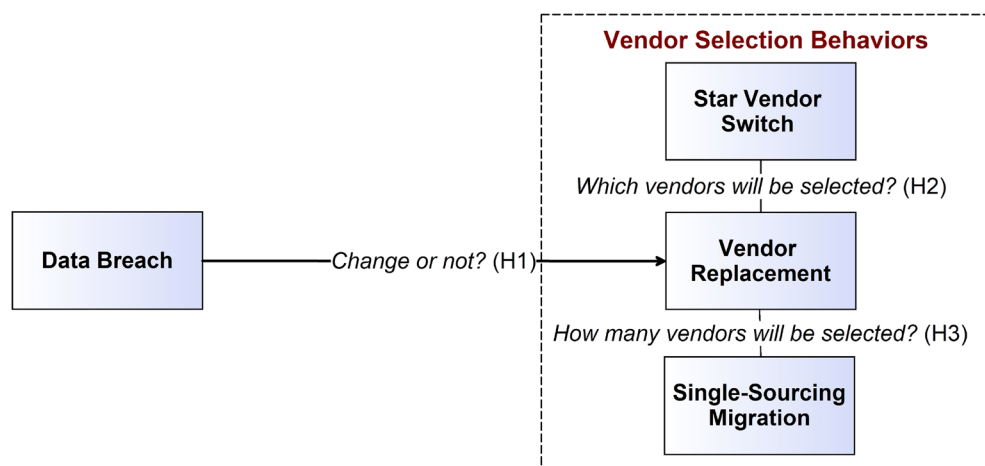
4.1 | Data and sample selection

To test our hypotheses, we collect data from multiple public sources. Healthcare data are obtained from the Healthcare Information and Management Systems Society (HIMSS), which provides health IT information for approximately 80% of US hospitals and has been widely used in previous research (e.g., Gardner et al., 2015; Wani & Malhotra, 2018). We primarily focus on HIMSS data regarding EMRS outsourcing. Additionally, we collect various hospital characteristics data from HIMSS, such as the number of licensed beds, net patient revenue, and expenses; these variables are used to construct control variables in our analysis. Because EMRS outsourcing involves multiple interconnected modules, we have followed the approach of prior literature (Angst et al., 2017; Ganju et al., 2022; Greenwood et al., 2019; Sharma et al., 2016) and focused on the five most commonly adopted EMRS modules as defined by HIMSS: Clinical Data Repository (CDR), Clinical Decision Support System (CDSS), Computerized Practitioner Order Entry (CPOE), Order Entry (OE), and Physician Documentation (PD). For more detailed descriptions of these modules, please refer to Appendix B.

We also collect data on data breaches in hospitals using two reliable sources, namely the Privacy Rights Clearinghouse (PRC) and the Identity Theft Resource Center (ITRC). This dual-source approach allows us to gather a wider range of information on data breaches. We conduct a thorough search of these sources to identify data breaches in hospitals that occur between 2006 and 2017. We manually compile all the reported breaches from the PRC by filtering the “Type of Organization” category to “Hospital” and from the ITRC by narrowing down the “Breach Category” to “Medical/Healthcare.” Additionally, we check the Health and Human Services (HHS) website for hospital data breaches to ensure completeness. While we do not find any new breaches from HHS, we obtain supplementary information about some breaches in our sample, such as breach city.

To mitigate concerns regarding endogeneity, our analysis primarily focuses on external breaches that are caused by external attackers. To achieve this, we manually differentiate between breaches resulting from external attacks and those originating from insiders within the hospitals. This distinction is made by referring to the breach descriptions provided by the PRC and ITRC (see Appendix C, Table C1). As a conservative approach, we exclude breaches without any descriptions initially. Subsequently, two authors, both experienced in

FIGURE 1 Conceptual framework.



information security studies, independently classify each breach as either an external attack or an internal incident based on the parties directly responsible for the data leakage, as documented in the breach descriptions (Cheng et al., 2017; Kwon & Johnson, 2014). There have been no disagreements between the two authors regarding the classification of breaches. By employing this methodology, we identify 2167 data breaches that stemming from external attacks and occur in hospitals across the US from 2006 to 2017.

We match this data breach sample (from PRC and ITRC) with healthcare data (from HIMSS) using hospitals' names and locations. We exclude observations with missing variables used in HIMSS. Consequently, we obtain a full sample comprising 60,793 observations from 5508 unique US hospitals. The sample covers a 12-year period from 2006 to 2017 and includes a total of 566 incidents of data breaches.

4.2 | Operationalization of variables

4.2.1 | Primary outcome variables

We investigate the relationship between data breaches and hospitals' vendor selection behaviors, focusing primarily on three outcome variables: (1) vendor replacement (*Replacement*), (2) switch to a star vendor (*Star Switch*), and (3) migration towards a single-sourcing configuration (*Single-Sourcing*). EMRS is an IT application that encompasses multiple modules, with each module being capable of assigning only one vendor at any given time. In order to effectively measure our dependent variables, we concentrate our investigation on the five most commonly adopted EMRS modules as designated by the HIMSS: CDR, CDSS, CPOE, OE, and PD (Angst et al., 2017).

Vendor replacement (Replacement)

For a specific module, if HIMSS records that a hospital uses a different vendor in the current year compared to the previous year, we consider that the hospital has replaced its vendor for that module in the current year. We operationalize *Replacement* for a hospital in a given year as a binary variable that equals 1 if the hospital replaces its vendor in any of the EMRS modules in a certain year, and 0 otherwise.

Switch to a star vendor (Star Switch)

For nearly 20 years, KLAS has recognized the best EMRS vendors in the US through its annual "Best in KLAS Awards" for Overall Software Suite Rankings.² The ranking methodology is based on a proprietary weighting system developed by the KLAS panel of experts, which integrates feedback from clinicians and managers from more than 4500 hospitals and 2500 outpatient clinics. We consider an EMRS vendor as a "star" if it has been listed as one of the top three in last year's KLAS annual ranking. Then, for a certain year in a chosen module, we define that a hospital switches to a star vendor in the module if the HIMSS records indicate that the hospital's vendor is not a star vendor in the last year but has been changed to a star vendor in the focal year. Then, we operationalize *Star Switch* as a binary variable that equals 1 if a hospital switches to a star vendor in any of the EMRS modules in a certain year, and 0 otherwise.

Migration towards a single-sourcing configuration (Single-Sourcing)

We define an EMRS single-sourcing configuration as a situation in which all EMRS modules have been designated to the same vendor. Then, we define the distance from a single-sourcing configuration as the *minimum* number of steps (i.e., times of module substitution) that a hospital has to take to transform its current EMRS

sourcing configuration into a single-sourcing one. Thereafter, we further construct *Single-Sourcing* as a binary variable that equals 1 if a hospital's distance from a single-sourcing configuration is reduced compared with the one in the last year, and 0 otherwise.

Table 1 illustrates the example calculations for *Replacement*, *Star Switch*, and *Single-Sourcing*.

4.2.2 | Controls

To account for the various factors that may affect a hospital's selection of EMRS vendors, we control for several hospital-level characteristics in our study. First, we incorporate *Hospital Size* as a control variable, represented by the number of licensed beds, as larger hospitals tend to have a lower likelihood of switching to a new EMRS provider (Kim et al., 2006). Second, we control for *Hospital Type*, as the choice of vendors should align with the specific type of hospital. Third, we include *Revenue* as a control variable, recognizing that a hospital's financial resources can impact their ability to switch vendors. Fourth, we incorporate *Marginal Profit* as a control variable, considering its potential influence on hospitals' budget allocation and their quality expectations during vendor selection processes. Fifth, the level of digitalization within hospitals determines the requirements for selecting EMRS vendors, and therefore, we control

for the *Digitalization Degree* of a hospital. Sixth, we account for a hospital's *IS Expense Intensity* as a control variable, as it has a notable impact on the financial and technological requirements associated with EMRS vendor selection. Lastly, we consider the operational environment of hospitals and control for the *System Size* of the healthcare system to which a hospital is affiliated. Definitions and data sources for all variables can be found in Appendix D.

4.3 | Identification strategy

We employ a DiD model to estimate the impact of data breaches on vendor selection behaviors. Only a subset of hospitals has experienced data breaches, providing us with an ideal setting to observe the effects of data breaches and mitigate unobserved confounders by differencing. However, data breaches are not random events; various factors, such as IT applications (McLeod & Dolezel, 2018; Wang et al., 2015) and IT governance (Haislip et al., 2021), influence the likelihood of a hospital experiencing a data breach. These same factors may also affect a hospital's EMRS vendor selection behaviors. To address this concern, we match each observation from a hospital that experiences a data breach with an observation from a hospital that has no data breach during our sample period. Our analysis is

TABLE 1 Example calculations for outcome variables.

Year	EMRS sourcing configuration (CDR-CDS-CPOE-OE-PD)	Operationalizing Replacement		Operationalizing Star Switch		Operationalizing Single-Sourcing		
		Module-level Replacement? (CDR-CDS-CPOE-OE-PD)	Variable value	Module-level Star Switch? (CDR-CDS-CPOE-OE-PD)	Variable value	Distance from Single-Sourcing configuration	Reduced compared with last year?	Variable value
2006	A-B-C*-D-D	Null	Null	Null	Null	Null	Null	Null
2007	A-B-C*-C*-D	No-No-No-Yes-No	1	No-No-No-Yes-No	1	3	No	0
2008	B-B-C*-C*-D	Yes-No-No-No-No	1	No-No-No-No-No	0	3	No	0
2009	B-B-C*-C*-C*	No-No-No-No-Yes	1	No-No-No-No-Yes	1	2	Yes	1
2010	B-B-C*-C*-C*	No-No-No-No-No	0	No-No-No-No-No	0	2	No	0
2011	B-C*-C*-C*-C*	No-Yes-No-No-No	1	No-Yes-No-No-No	1	1	Yes	1
2012	B-C*-C*-C*-C*	No-No-No-No-No	0	No-No-No-No-No	0	1	No	0
2013	B-C*-C*-C*-C*	No-No-No-No-No	0	No-No-No-No-No	0	1	No	0
2014	C*-C*-C*-C*-C*	Yes-No-No-No-No	1	Yes-No-No-No-No	1	0	Yes	1
2015	C*-C*-C*-C*-C*	No-No-No-No-No	0	No-No-No-No-No	0	0	No	0
2016	C*-C*-C*-C*-C*	No-No-No-No-No	0	No-No-No-No-No	0	0	No	0
2017	C*-C*-C*-C*-C*	No-No-No-No-No	0	No-No-No-No-No	0	0	No	0

Note: We suppose Vendor C is a star EMRS vendor and use * to indicate this role of the vendor. The highlighted numbers are intended to emphasize the construction of our dependent variable

conducted using this matched sample to account for potential confounding variables.

To construct matched samples, we employ the CEM method developed by Rosenbaum and Rubin (1983). CEM is a non-parametric technique designed to improve comparability between groups by making the distributions of covariates as similar as possible. By achieving balance on covariates, CEM helps address confounding factors in observational data, thereby enhancing the validity of causal inference in observational studies. We apply the CEM method to our full sample consisting of 60,793 observations. Our CEM approach employ a many-to-many matching strategy. All continuous control variables are selected as pre-treatment covariates in the matching process. Additionally, *Hospital Size*, *Revenue*, and *Profit Margin* are coarsened based on their 10th, 25th, 50th, 75th, and 90th percentiles. Moreover, we apply the Freedman-Diaconis rule to ensure that the distribution of the matched control observations closely resembles that of the treatment group. After matching each treatment observation to a control, we remove the matched control observation to ensure that each treatment observation is only matched with one control observation. Finally, we exclude unmatched observations from our DiD analysis and obtain a 1:1 matched treatment and control groups, each consisting of 5156 observations. Thus, the total CEM-matched sample includes 10,312 observations.

Table 2 indicates the effectiveness of the CEM. The overall imbalance measure (multivariate *L1* distance),³ as shown in Table 2, exhibits a significant decrease from 0.905 to 0.611 after matching. Furthermore, the imbalance for each covariate is notably reduced compared to before matching.

After identifying the comparable treatment and control groups using CEM, we apply the following DiD specification:

TABLE 2 Variable imbalance (*L1* distance) comparison before and after coarsened exact matching.

Variable	Before matching	After matching
<i>Hospital Size</i>	0.334	0.107
<i>Revenue</i>	0.349	0.107
<i>Profit Margin</i>	0.050	0.024
<i>IS Expense Intensity</i>	0.066	0.036
<i>System Size</i>	0.123	0.021
Multivariate <i>L1</i> distance	0.905	0.611

Note: The highlighted numbers represent the indicators for the success of this test. Bold indicates important terms.

$$\text{Outcomes}_{i,j} = \beta_0 + \beta_1 \text{After Breach}_{i,j} \times \text{Breach}_i \quad (1) \\ + \sum \alpha \text{Control}_{i,j} + v_i + w_j + \epsilon_{ij}.$$

Here, subscripts *i* and *j* represent hospital and year, respectively. The outcome variable, *Outcome*, represents three variables, namely, vendor replacement (*Replacement*), switch to a star vendor (*Star Switch*), and single-sourcing migration (*Single-Sourcing*). *After Breach* is a time-variant indicator that equals 1 for the years that after the first data breach during our sample period, and 0 otherwise. *Breach* is a treatment group indicator that equals 1 for the observations in the treatment group and 0 otherwise. *Control* represents a series of control variables that we consider, and α denotes the corresponding estimated coefficient. *v* represents hospital-fixed effects, controlling for any time-invariant unobserved confounders, while *w* represents year-fixed effects, controlling for systemic differences that may influence hospital EMRS vendor selection. Notably, the separate terms for *Breach* and *After Breach* do not appear in the model because the former is absorbed by the hospital-fixed effects and the latter is collinear with the interaction term between the two. Lastly, the error term is represented by ϵ .

Our identification strategy should rely on two key assumptions, namely, parallel pre-trends and the exogeneity of treatment. We will provide specific evidence for these assumptions in Section 5.3.

5 | EMPIRICAL RESULTS

After merging data from multiple sources and applying the CEM technique, we obtain a CEM-matched sample comprising 10,312 hospital-year observations from 2006 to 2017. Our subsequent DiD analysis will be based on this sample.

5.1 | Descriptive statistics

Table 3 provides the descriptive statistics of the variables used in the subsequent DiD analysis for the overall CEM-matched sample, as well as three specific groups: (1) treatment hospitals before the breach, which includes 2743 observations, (2) treatment hospitals after the breach, which includes 2413 observations, and (3) control hospitals, which includes 5156 observations. Table 3 indicates that approximately 23.4% of the hospital-year observations are affected by data breaches. In addition, within the overall CEM-matched sample, the average value of *Breach* is 0.5, indicating that the treatment and control groups have an equal number of observations.

Table 3 provides insights into the trends in hospital EMRS vendor selection over the study period. The table reveals that approximately 15.2% of hospitals replace at least one EMRS vendor annually. Among these hospitals, about 35.5% (5.4%/15.2%) choose to switch to a star vendor, while approximately 33.6% (5.1%/15.2%) further migrate towards a single-sourcing configuration. Obviously, these findings indicate that during the focused period, the prevailing trends in hospitals' EMRS vendor selection involve switching to non-star (e.g., startup) vendors and migrating towards a "best-of-breed" (i.e., multi-sourcing) arrangement.

Table 3 also provides model-independent estimates regarding hospitals' high propensity to replace vendors, switch to star vendors, and migrate to single-sourcing after data breaches. Specifically, in terms of vendor replacement, the average probability of vendor replacement in treatment hospitals before a breach is 13.5%, which significantly increases to 17.7% in treatment hospitals after a breach. In contrast, control hospitals have an average probability of vendor replacement of 14.8%. Regarding the probability of switching to a star vendor, the probability among treatment hospitals prior to a breach is 3.6%, but significantly increases to 7.4% among treatment hospitals after a breach. In contrast, control hospitals have an average probability of 5.3% of switching to a star

vendor. Similarly, when considering the probability of single-sourcing migration, among treatment hospitals prior to a breach, the average probability is 5.0%. However, this probability increases to 5.2% after the occurrence of a data breach. In contrast, control hospitals have an average probability of 5.1% of migrating to a single-sourcing configuration.

5.2 | Baseline results

Table 4 presents the results of our baseline DiD regression analysis, which examines the impact of data breaches on EMRS vendor selection behavior. Due to the binary nature of the outcome variables and the longitudinal nature of the data, we employ a fixed-effects *logit* model. To account for potential correlation of regression residuals across years for a given hospital, we cluster robust standard errors at the hospital level. Additionally, in the robustness check, we alternatively use a fixed-effects linear probability model (LPM).

Models 1–3, Table 4, only include the double-difference term (*After Breach* $\times$ *Breach*) as the sole independent variable. In Model 1, the dependent variable is *Replacement*. The use of a fixed-effects logit model automatically excludes all the hospitals that have no change in *Replacement* across the years from our sample. This

TABLE 3 Summary statistics.

Variables	All observations				Treatment group				Control group	
					Before Breach		After Breach			
	Mean	SD	Min	Max	Mean	SD	Mean	SD	Mean	SD
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Replacement</i>	0.152	0.359	0.000	1.000	0.135	0.342	0.177	0.382	0.148	0.355
<i>Star Switch</i>	0.054	0.225	0.000	1.000	0.036	0.187	0.074	0.262	0.053	0.224
<i>Single-Sourcing</i>	0.051	0.220	0.000	1.000	0.050	0.217	0.052	0.223	0.051	0.221
<i>After Breach</i>	0.234	0.423	0.000	1.000	0.000	0.000	1.000	0.000	0.000	0.000
<i>Breach</i>	0.500	0.500	0.000	1.000	1.000	0.000	1.000	0.000	0.000	0.000
<i>Hospital Size</i>	5.187	1.021	2.303	6.631	5.071	1.061	5.340	1.009	5.177	0.996
<i>Hospital Type</i>	6.921	3.088	1.000	18.000	6.696	3.194	6.683	3.571	7.153	2.751
<i>Revenue</i>	3.370	3.949	−0.818	50.814	2.887	3.377	4.620	5.225	3.041	3.374
<i>Profit Margin</i>	−0.146	1.342	−11.544	0.423	−0.039	0.851	−0.328	1.882	−0.119	1.240
<i>Digitalization Degree</i>	2.119	1.809	0.000	4.000	1.280	1.697	2.968	1.507	2.169	1.799
<i>IS Expense Intensity</i>	0.004	0.013	0.000	0.170	0.003	0.011	0.005	0.015	0.003	0.012
<i>System Size</i>	1.815	1.222	0.693	5.313	1.692	1.182	1.953	1.252	1.816	1.222

Note: This table presents the summary statistics of the main variables. All variables are defined in Appendix D. Columns (1)–(4) present descriptive statistics for the overall CEM-matched sample. Columns (5) and (6) report the mean and standard deviation of observations in the treatment group in or before the year of the first data breach. Columns (7) and (8) report the mean and standard deviation of observations in the treatment group after the year of the first data breach. Columns (9) and (10) report the mean and standard deviation of observations in the control group.

TABLE 4 Impact of data breach on vendor selection behaviors.

	<i>Replacement</i>	<i>Star Switch</i>	<i>Single-Sourcing</i>	<i>Replacement</i>	<i>Star Switch</i>	<i>Single-Sourcing</i>
Variables	(1)	(2)	(3)	(4)	(5)	(6)
<i>After Breach</i> × <i>Breach</i>	0.360*** (3.01)	0.576*** (2.74)	0.670*** (3.48)	0.344*** (2.86)	0.521** (2.46)	0.619*** (3.20)
<i>Hospital Size</i>				0.257 (1.00)	−0.116 (−0.26)	0.344 (0.83)
<i>Revenue</i>				−0.004 (−0.18)	−0.005 (−0.15)	−0.074** (−2.14)
<i>Profit Margin</i>				−0.008 (−0.27)	−0.030 (−0.51)	0.182** (2.42)
<i>IS Expense Intensity</i>				8.201*** (2.83)	3.039 (0.59)	9.943* (1.86)
<i>System Size</i>				0.071 (0.83)	0.147 (0.94)	0.201 (1.22)
Hospital-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	6129	3437	3221	6129	3437	3221
Pseudo R ²	.069	.073	.101	.072	.077	.113

Note: This table presents coefficients from DiD regressions of the effect of data breaches on vendor selection behaviors. We use a fixed-effects logit model. The sample is a CEM-matched sample, which includes 10,312 observations. In each regression, the application of a fixed-effects logit model automatically excludes hospitals that do not show any variation in the corresponding outcome variable across years. Models 4–6 control for *Hospital Type* and *Digitalization Degree*. Standard errors are clustered by hospital. *t* statistics are in parentheses (**p* < .1; ***p* < .05; ****p* < .01). The highlighted numbers are intended to emphasize the construction of our dependent variable

leads to the exclusion of 4183 observations, leaving us with 6129 (10,312 − 4183) observations for analysis. In this model, the estimated coefficient for the double-difference term is 0.360 (significant at the 1% level). This suggests that among hospitals that replaced EMRS vendors during the sample period, treatment hospitals (vs. control hospitals) are more likely to replace EMRS vendors after a data breach, providing support for H1.

In Model 2, Table 4, the dependent variable is *Star Switch*. The use of fixed-effects logit model excludes hospitals that did not show any change in *Star Switch* across the years from our sample. As a result, 6295 observations are excluded, leaving us with 3437 (10,312 − 6295) observations for analysis. In this model, the estimated coefficient for the double-difference term is 0.576 (significant at the 1% level). This suggests that among hospitals that switched towards a star vendor during the sample period, treatment hospitals (vs. control hospitals) are more likely to switch to star vendors after a data breach, providing support for H2.

In Model 3, Table 4, the dependent variable is *Single-Sourcing*. The fixed-effects logit model excludes hospitals that do not exhibit any change in *Single-Sourcing* from our sample. As a result, 7091 observations are excluded,

leaving us with 3221 (10,312 − 7091) observations for analysis. In this model, the estimated coefficient for the double-difference term is 0.670 (significant at the 1% level). This indicates that among hospitals that migrated towards a single-sourcing configuration during the sample period, treatment hospitals (vs. control hospitals) are more likely to further migrate towards a single-sourcing configuration after a data breach, providing support for H3.

In Models 4–6, Table 4, we have introduced all control variables into the models. Notably, the coefficients for the double-difference terms remain statistically significant and hold economic significance. To facilitate comparisons, we converted these coefficients into odds ratios. Specifically, in Model 4, the coefficient for the double-difference term is 0.344, corresponding to an odds ratio of 1.410. This implies that among hospitals that replaced EMRS vendors during the sample period, the odds of replacing EMRS vendors for treatment hospitals (vs. control hospitals) are 1.410 times after a data breach over before it. In Model 5, the coefficient for the double-difference term is 0.521, corresponding to an odds ratio of 1.684. This implies that among hospitals that switched towards a star vendor during the sample period, the odds

of switching to a star vendor for treatment hospitals (vs. control hospitals) are 1.684 times after a data breach over before it. Furthermore, in Model 6, the coefficient for the double-difference term is 0.619, with a corresponding odds ratio of 1.856. This suggests that among hospitals that migrated towards a single-sourcing configuration during the sample period, the odds of migrating towards single-sourcing for treatment hospitals (vs. control hospitals) are 1.856 times after a data breach over before it.

5.3 | Identification checks

The estimated results above indicate that, after data breaches, hospitals tend to replace vendors, switch to star vendors, and migrate towards single-sourcing. To identify these effects as causal, we inspect the two critical assumptions that underlie the DiD approach for causal interpretation: parallel pre-trends and the exogeneity of treatment.

5.3.1 | Testing for parallel pre-trends

The assumption of parallel pre-trends means that in the absence of data breaches, the trends of the outcome variables of interest (vendor replacement, selection of star vendors, and single-sourcing migration) are parallel between the treatment group and the control group. If significant differences in these trends exist, it violates the parallel pre-trends assumption, suggesting the presence of potential confounding factors that may confound the causal interpretation of the data breach effect. To test this assumption, a common approach is to examine whether the trends of the outcome variables are parallel between the treatment group and the control group before the implementation of the treatment (i.e., data breaches in our study). To test for parallel pre-trends, we conduct a dynamic DiD test to examine how the impact of data breaches vary over time. We run similar models to the baseline model specification as follows:

$$\text{Outcomes}_{i,j} = \beta_0 + \sum_{-5 \leq t \leq 5} \beta_t \text{Year}_{i,j}^t \times \text{Breach}_i \quad (2) \\ + \sum \alpha_r \text{Control}_{i,j} + v_i + w_j + \epsilon_{ij}.$$

In this equation, t represents the number of years relative to the year of the data breach (e.g., if a hospital experiences a data breach in 2010, then $t = -1$ for the year 2009 and $t = 1$ for the year 2011). Year^t is defined as an indicator that equals 1 if the year corresponds to the

relative t -th year from the hospital's first data breach. To avoid multicollinearity, we follow the standard practice of dynamic DiD and exclude the period closest to the data breach before the treatment (i.e., Year^{-1}) (e.g., Baker et al., 2022; Zhang et al., 2023).

Table 5 presents the results of the dynamic DiD test. In Table 5, for each outcome variable, the coefficients for the pre-treatment period are nearly insignificant, indicating that the parallel pre-trends assumption is satisfied. Conversely, the coefficients for the post-treatment periods become positive and significant for each outcome variable, suggesting that hospitals tend to respond *immediately* after data breaches by replacing vendors, switching to star vendors, and migrating towards single-sourcing. Additionally, the difference in the significance between the lagged and lead indicators demonstrates that the results are not influenced by reverse causality.

Figure 2 further illustrates the results of the dynamic DiD test along with the 90% confidence intervals. For $t < 0$ (i.e., the pre-treatment years), the point estimates of β_t are centered around the zero level. This further confirms that the parallel pre-trends assumption of the DiD analysis is not violated, and our results are not driven by pre-treatment differences in trends between the treatment and control groups.

5.3.2 | Testing for exogeneity of treatment

The second assumption is the exogeneity of treatment, which assumes that the occurrence of data breaches is not influenced by endogenous factors related to hospital vendor behavior. Violation of this assumption could result in the confounding effect on the identification of causal relationships.

We have tackled the issue of endogenous treatment from various perspectives. For instance, we incorporate hospital-fixed effects to control for time-invariant confounders that could potentially affect vendor selection behaviors. We also focus on external breaches only in our models to minimize the possibility of internal attackers endogenously selecting the timing or targets of attacks. Additionally, inspired by Chu et al. (2021) and Bertrand et al. (2004), we conduct a placebo test. Previous research suggests that placebo tests can provide evidence in support of unconfoundedness and “as-if” random design (De la Cuesta & Imai, 2016; Hartman & Hidalgo, 2018). We adopt the placebo test method proposed by La Ferrara et al. (2012), which involves randomly assigning data breach dates to non-breached hospitals to construct a randomized experiment. A more detailed description of this testing process can be found in Appendix E. Figure 3

TABLE 5 Dynamic DiD test.

Variables	<i>Replacement</i> (1)	<i>Star Switch</i> (2)	<i>Single-Sourcing</i> (3)
$Year^{-5} \times Breach$	−0.128 (−0.58)	−0.397 (−1.02)	0.325 (0.97)
$Year^{-4} \times Breach$	0.204 (0.94)	0.049 (0.13)	0.366 (1.08)
$Year^{-3} \times Breach$	−0.151 (−0.72)	−0.373 (−0.93)	0.025 (0.07)
$Year^{-2} \times Breach$	0.244 (1.41)	0.225 (0.79)	−0.048 (−0.16)
<i>Current</i>	−0.279 (−1.64)	−0.430 (−1.48)	−0.359 (−1.25)
$Year^1 \times Breach$	0.328** (2.05)	0.508** (2.17)	0.481* (1.95)
$Year^2 \times Breach$	0.106 (0.61)	0.353 (1.44)	0.245 (0.91)
$Year^3 \times Breach$	0.153 (0.89)	0.044 (0.17)	0.164 (0.56)
$Year^4 \times Breach$	0.007 (0.04)	0.128 (0.46)	−0.127 (−0.36)
$Year^5 \times Breach$	0.372** (2.15)	0.128 (0.49)	0.139 (0.40)
Controls	Yes	Yes	Yes
Hospital-fixed effects	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes
Number of observations	6129	3437	3221
Pseudo R^2	.071	.078	.101

Note: This table presents results of dynamic DiD test. We use a fixed-effects logit model. The sample is a CEM-matched sample, which includes 10,312 observations. In each regression, the application of a fixed-effects logit model automatically excludes hospitals that do not show any variation in the corresponding outcome variable across years. Standard errors are clustered by hospital. t statistics are in parentheses ($*p < .1$; $**p < .05$; $***p < .01$). The highlighted numbers are intended to emphasize the construction of our dependent variable

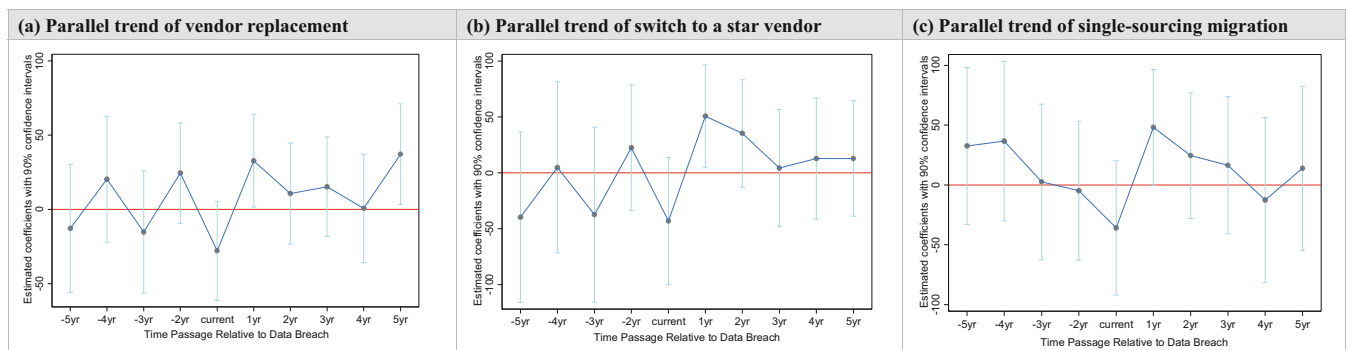


FIGURE 2 Parallel trends plot.

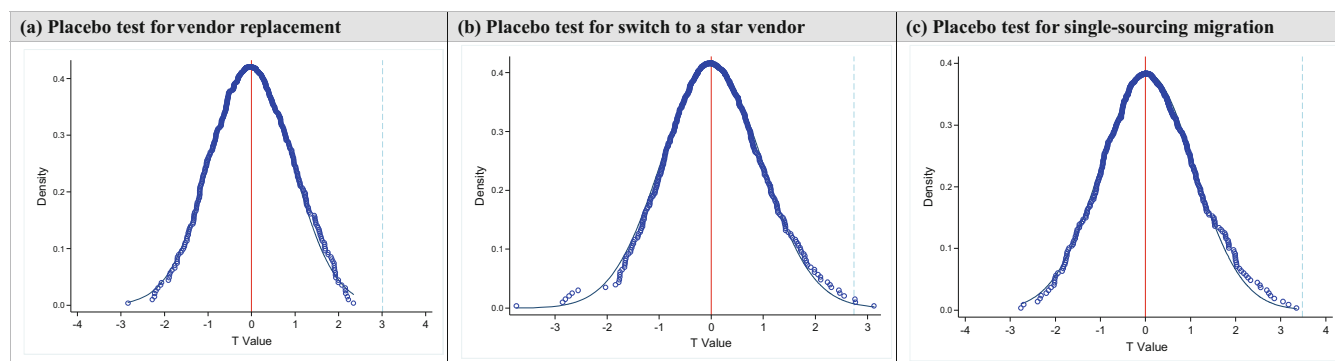


FIGURE 3 Placebo test.

displays the results of the placebo test and shows that the t -values of the pseudo-double-difference terms for each outcome variable are predominantly centered around zero, consistent with the fact that fake data breach dates can be viewed as classical measurement errors.

5.4 | Robustness checks

In this section, we have outlined a number of additional concerns that may introduce bias into our estimation, along with the robustness check that we conduct to address each concern.

5.4.1 | Entropy balancing matching

CEM matching can lead to a substantial loss of sample data during the matching process, potentially weakening the persuasiveness of the results. Additionally, our results may be subject to bias introduced by the matching method. To alleviate these concerns, we employ an alternative matching technique of entropy balancing, which is a novel multivariate matching approach. Entropy balancing, introduced by Hainmueller (2012), has been widely adopted in recent studies (e.g., Hendricks et al., 2019; Zhang et al., 2023). This method allows for reweighting the dataset, enabling the creation of a control group sample with covariate distributions similar to the treatment group. This helps to achieve more parallel pre-trends in our DiD analysis without sacrificing sample size (Hainmueller, 2012).

We use entropy balancing matching as an alternative to CEM in our full sample to create matched samples. All continuous control variables are specified as covariates to be balanced using entropy balancing. The results of the entropy balancing process are presented in Appendix F, which support the effectiveness of this method.

We repeat the DiD estimation in Models 4–6 of Table 4 on the entropy balancing-matched sample. The fixed-

effects LPM model is utilized to incorporate the weights in our analysis.⁴ The results of this analysis are presented in Columns (1)–(3) of Table 6. In these columns, the double-difference terms are all statistically significant, suggesting that hospitals are more inclined to replace vendors, switch to star vendors, and further migrate towards single-sourcing after data breaches. These results provide further evidence of the robustness of our findings.

5.4.2 | Addressing inequality in modules

Our previous analysis is at the hospital–year level, in which we treat the five EMRS modules (i.e., CDR, CDSS, CPOE, OE, or PD) equally. Such a design may suffer from the concern that some modules (e.g., PD) are more important and less likely to be replaced. To mitigate this concern, we further provide the analyses at the hospital–module–year level. We show the results of the analysis in Columns (4)–(6), Table 6. The results are consistent with the baseline analysis and further support the robustness of our findings.

5.4.3 | Alternative model

Another concern is whether our results are sensitive to the model specification. To address this, we replicate our baseline analyses using a fixed-effects LPM model alternatively. Columns (7)–(9), Table 6, show that the results are entirely consistent with the baseline analysis, thus alleviating this concern.

6 | POST-HOC INVESTIGATION: EXPLORING CONTINGENCY FACTORS

Previous sections demonstrate that hospitals tend to change their EMRS vendors in the wake of data breaches,

TABLE 6 Robustness checks.

Variables	Entropy balancing matching			Additional module-fixed effects			Fixed effect LPM regressions		
	Replacement	Star Switch	Single-Sourcing	Replacement	Star Switch	Single-Sourcing	Replacement	Star Switch	Single-Sourcing
<i>After Breach</i> × <i>Breach</i>	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	0.040** (2.14)	0.031** (2.29)	0.029*** (2.62)	0.347*** (2.87)	0.527** (2.45)	0.599*** (3.11)	0.044*** (2.89)	0.026*** (2.60)	0.029*** (3.09)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Module fixed effects	No	No	No	Yes	Yes	Yes	No	No	No
Number of observations	60,793	60,793	60,793	6129	3437	3221	10,312	10,312	10,312

Note: This table presents the coefficients obtained from DiD regressions that examine the impact of data breaches on vendor selection behaviors (robustness checks). In Columns (1)–(3), we replicate Model 4–6 in Table 4 by alternatively employing the entropy balancing-matched sample and applying fixed-effects LPM. In Columns (4)–(6), we replicate Model 4–6 in Table 4 by adding the module-fixed effects. In Columns (7)–(9), we replicate Model 4–6 in Table 4 by changing the fixed-effects logit model to the fixed-effects LPM. Standard errors are clustered by hospital. *t* statistics are in parentheses (**p* < .1; ***p* < .05; ****p* < .01). The highlighted numbers are intended to emphasize the construction of our dependent variable

frequently involving the manner of switching to star vendors or migrating towards a single-sourcing configuration. This section further infers the potential generalizability of the findings by examining how the effects of data breaches are contingent upon different factors.

6.1 | How does client–vendor relationship duration matter?

We first consider the contingency effect of client–vendor relationship duration. We argue that as the duration of the relationship with an EMRS vendor increases, client hospitals tend to exhibit a stronger resistance to vendor replacement following data breaches. The reason is that a long-standing client–vendor relationship fosters familiarity, trust, and positive expectations regarding the future behavior of their vendors (Claycomb & Frankwick, 2004; Dyer & Chu, 2000; Vanneste et al., 2014). A prolonged relationship also facilitates the development of relationship-specific expertise and knowledge and generates routines that are relevant only within the context of the current relationship (Kim et al., 2006). The accumulation of familiarity and highly relationship-specific knowledge poses challenges for employees in adapting to new vendors (Yi et al., 2016), thereby increasing the difficulties faced by breached hospitals in establishing new relationships. From the ABV perspective, these difficulties will consume decision-makers' attention, thus reducing the attention and effort allocated to vendor replacement (Ocasio, 1997; Sullivan, 2010). Accordingly, we predict that the impact of data breaches on (a) EMRS vendor replacement, (b) star EMRS vendor switch, or (c) EMRS single-sourcing migration would be weakened as the client–vendor relationship duration increases.

To construct client–vendor relationship duration (*Duration*), we first measure the module-level relationship duration of a specific EMRS module as the number of years since the hospital signed the contract with the vendor to the end of the current year. Then, at the hospital level, we operationalize *Duration* as the maximum value of five module-level relationship durations. Columns (1)–(3), Table 7, present the coefficients from DiD regressions of the moderating role of relationship duration (*Duration*). Column (1) shows that the coefficient of *After Breach* × *Breach* × *Duration* is −0.222 and significant at a 1% level, indicating that the effect of data breach on vendor replacement attenuates as the relationships between client hospitals and EMRS vendors last long. Column (2) exhibits a consistent and negative coefficient of *After Breach* × *Breach* × *Duration*; however, the coefficient is insignificant. Column (3) shows that the coefficient of *After Breach* × *Breach* × *Duration* is

TABLE 7 Moderating role of relationship duration and hospital system size.

Variables	Star		Single-Sourcing		Star		Single-Sourcing		Star		Star	
	Replacement	Switch	Replacement	Switch	Replacement	Switch	Replacement	Switch	Replacement	Switch	Replacement	Switch
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
<i>After Breach</i> × <i>Breach</i>	0.703*** (3.84)	0.580* (1.75)	1.060*** (3.34)	0.750*** (3.50)	1.474*** (3.60)	0.948*** (2.58)	1.128*** (4.42)	1.552*** (3.29)	1.413*** (3.19)			
<i>After Breach</i> × <i>Breach</i> × <i>Duration</i>	−0.222*** (−2.60)	−0.050 (−0.33)	−0.316** (−2.02)				−0.231*** (−2.70)	−0.053 (−0.34)	−0.315** (−2.00)			
<i>After Breach</i> × <i>Breach</i> × <i>System Size</i>				−0.199** (−2.25)	−0.508*** (−2.84)	−0.178 (−1.22)	−0.208** (−2.37)	−0.508*** (−2.85)	−0.174 (−1.23)			
Moderators	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	6129	3437	3221	6129	3437	3221	6129	3437	3221	3437	3221	3221
Pseudo R ²	.121	.152	.183	.121	.143	.186	.124	.152	.187			

Note: This table presents coefficients from DiD regressions of the moderating effect of relationship duration and hospital system size. We use a fixed-effects logit model. The sample is a CEM-matched sample, which includes 10,312 observations. In each regression, the application of a fixed-effects logit model automatically excludes hospitals that do not show any variation in the corresponding outcome variable across years. Standard errors are clustered by hospital. *t* statistics are in parentheses (**p* < .1; ***p* < .05; ****p* < .01). The highlighted numbers are intended to emphasize the construction of our dependent variable

significantly negative ($\beta = -.316, p < .05$), suggesting that a longer relationship with the existing vendors weakens the driving forces of data breach on single-sourcing migration. Panel A of Figure 4 visually demonstrates the moderating effect of *Duration* on vendor selection behaviors by using marginal effect plots, following the approach outlined in Brambor et al. (2006). Panel A of Figure 4 illustrates that in the presence of a long-standing relationship, the positive marginal effect of a data breach on each of the following outcomes—vendor replacement, star vendor switch, or single-sourcing migration—is weakened. This graphical representation aligns with the findings from our regression models and provides evidence supporting the weakening moderating effects of client–vendor relationship duration on the main effects.

6.2 | How does hospital system size matter?

We also focus on the contingency effect of hospital system size, which refers to the total number of hospitals within a multi-hospital healthcare system. Currently, over 70% of US hospitals belong to multi-hospital healthcare systems (Furukawa et al., 2020). In the context of a

large healthcare system, the operations of a hospital depend on its sister hospitals. This dependence is attributed to the extensive communication and collaboration that takes place among sister hospitals, enabling the development of shared expertise and processes (e.g., Li et al., 2022; Vest et al., 2019). Consequently, hospitals within a large healthcare system are more locked into existing relationships compared to independent hospitals (Kim et al., 2006). For instance, in a large multi-hospital system, a hospital often needs to establish information exchange connections with multiple sister hospitals to share patient information stored in the EMRS (e.g., Li et al., 2022; Vest et al., 2019). However, if an individual hospital within the system decides to replace its EMRS vendor, it may encounter technical challenges arising from the use of incompatible data storage formats introduced by a different vendor. From the ABV perspective, the aforementioned difficulties may further consume decision-makers' attention that is subsequently afforded to vendor replacement efforts. Therefore, we conjecture that the impact of data breaches on (a) EMRS vendor replacement, (b) star EMRS vendor switch, or (c) EMRS single-sourcing migration weakens as hospital system size increases.

We measure the hospital system size (*System Size*), which also serves as a control variable in our study, by

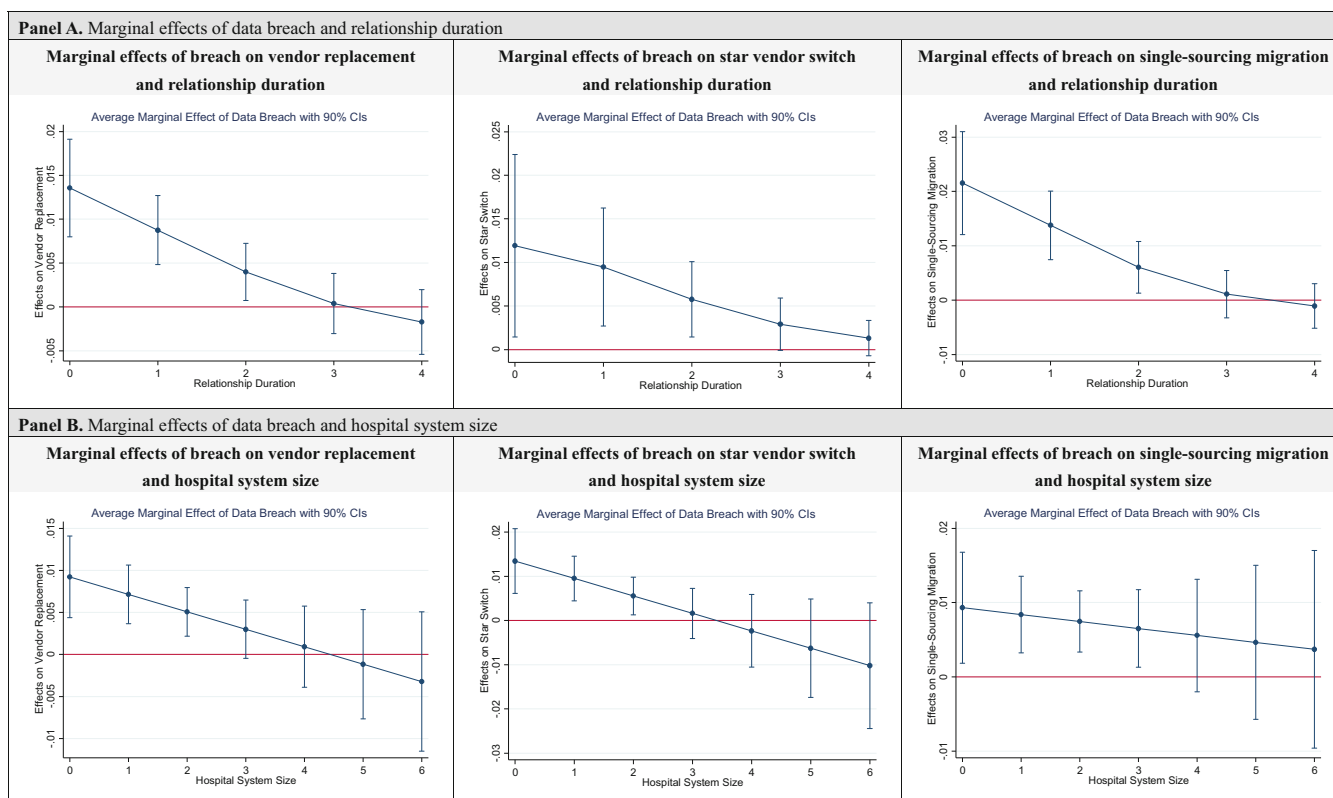


FIGURE 4 Marginal effect plot.

counting the total number of affiliated hospitals within the parent healthcare group of the focal hospital. Subsequently, we take the logarithm of the total number. Columns (4)–(6), Table 7, present the results of the DiD regression analysis, examining the moderating role of hospital system size (*System Size*). In Column (4), the coefficient of *After Breach* $\times$ *Breach* $\times$ *System Size* is significantly negative ($\beta = -.199$, $p < .05$), suggesting that the impact of a data breach on vendor replacement in breached hospitals is attenuated when they are part of a large hospital system. Similarly, in Column (5), the coefficient of *After Breach* $\times$ *Breach* $\times$ *System Size* is significant and negative ($\beta = -.508$, $p < .01$), indicating that the effect of data breaches on star vendor switching is weakened in breached hospitals belonging to large hospital systems. Furthermore, in Column (6) exhibits a consistent and negative coefficient of *After Breach* $\times$ *Breach* $\times$ *System Size*; however, the coefficient is insignificant. To facilitate a clear interpretation of these findings, we have employed a marginal effect plot to illustrate the moderating effect of *System Size* in Panel B of Figure 4. This panel visually shows that a large hospital system size weakens the positive marginal effect of a data breach on each of the following outcomes: vendor replacement, star vendor switch, and single-sourcing migration. This graphical representation aligns with the findings obtained from the regression models and provides evidence of the weakening moderating effects of hospital system size on the main effects.

We also examine *Duration* and *System Size* in the same model to assess their respective effects. The results, presented in Columns (7)–(9) of Table 7, are largely consistent with the earlier results and provide additional support for the robustness of our findings.

6.3 | How does breach scale matter?

After establishing the impact of data breaches on hospitals' vendor selection behavior, we delve into the role of breach scale in this process. Breach scale refers to the number of compromised records in a data breach. Larger-scale data breaches tend to garner more attention to information security, as evidenced by the positive correlation between the number of affected records and data breach costs (Ponemon Institute, 2014) as well as customer confidence loss (Garg et al., 2003; Kwon & Johnson, 2015). Consequently, we anticipate a more pronounced relationship between data breaches and vendor selection following larger-scale data breaches.

Additionally, we are interested in understanding whether the impact of a data breach on vendor behavior is long-term or short-term. Following a data breach,

hospitals typically respond quickly and intensify their focus on information security. However, over time, hospitals may gradually shift their attention away from data security towards other pressing or emergent matters. Consequently, long-term interference from other factors may diminish the effect of the data breach. Therefore, we expect the impact of a data breach to be short-term rather than long-term.

To test the above conjectures, we examine the dynamic effects of data breaches on vendor selection behavior, varying according to breach scale and the period around the breach.

The results, presented in Table 8, yielded the following insights. First, the effect of data breaches on hospitals' vendor replacement is contingent upon the scale of the breach, with the effect disappearing for breach scales below approximately 1000. Furthermore, this effect is short-term, as it materializes in the year of the data breach, prompting hospitals to promptly replace vendors, and disappears in the following year. Second, similar to vendor replacement, the impact of data breaches on hospitals' switch to star vendors is also contingent upon breach scale and exhibits a short-term nature. The effect disappears for breach scales below approximately 10,000, and it becomes evident immediately in the year of the breach, subsequently fading away. Moreover, we observe that the impact of data breaches driving hospitals to migrate towards a single-sourcing configuration is also dependent on breach scale and displays a short-term nature. Specifically, the effect weakens when the breach scale is less than 1000 or after the breach year.

In conclusion, we find that the impact of data breaches on hospitals' vendor selection behavior is short-term, with the significance of the effect often contingent upon the breach scale.

7 | DISCUSSION AND IMPLICATIONS

Vendor selection is a critical topic in the field of OM. In the digital age, where big data and cloud computing technology are widely used, organizations and vendors often share information and data, making information security a vital concern in selecting third-party vendors. However, the impact of information security on vendor selection lacks empirical evidence. This paper aims to fill this research gap by exploring how data breaches affect hospitals' vendor selection behavior from an ABV perspective. Our empirical analysis shows that hospitals tend to change their EMRS vendors following data breaches,

TABLE 8 Coefficient matrix for varying breach scale and periods around the breaches.

Variables	Vendor Replacement				Star Switch				Single-Sourcing			
	Year	Current	Year	Year	Year	Current	Year	Year	Year	Current	Year	Year
	<i>t</i> − 1	(2)	<i>t</i> + 1	<i>t</i> + 2	<i>t</i> − 1	(6)	<i>t</i> + 1	<i>t</i> + 2	<i>t</i> − 1	(10)	<i>t</i> + 1	<i>t</i> + 2
	(1)		(3)	(4)	(5)		(7)	(8)	(9)		(11)	(12)
<i>After Breach_100 × Breach</i>	0.363 (1.61)	0.274 (1.42)	−0.143 (−0.66)	−0.085 (−0.30)	0.529 (1.47)	0.498 (1.45)	0.023 (0.07)	−0.595 (−1.39)	0.366 (0.85)	0.569 (1.43)	0.117 (0.29)	−0.473 (−1.05)
<i>After Breach_1000 × Breach</i>	0.167 (1.00)	0.250* (1.72)	0.013 (0.09)	0.171 (0.94)	0.100 (0.37)	0.175 (0.67)	−0.141 (−0.55)	−0.259 (−0.86)	0.392 (1.30)	0.580** (2.05)	0.264 (0.92)	0.028 (0.09)
<i>After Breach_10,000 × Breach</i>	0.120 (0.91)	0.319*** (2.59)	0.144 (1.15)	0.177 (1.28)	0.241 (1.07)	0.522** (2.33)	0.163 (0.77)	−0.202 (−0.87)	0.337 (1.56)	0.545*** (2.65)	0.320 (1.53)	0.036 (0.15)
<i>After Breach_100,000 × Breach</i>	0.134 (1.05)	0.335*** (2.79)	0.119 (0.98)	0.115 (0.90)	0.263 (1.25)	0.492** (2.31)	0.164 (0.81)	−0.217 (−1.00)	0.358* (1.74)	0.656*** (3.34)	0.242 (1.22)	0.034 (0.15)
<i>After Breach_1,000,000 × Breach</i>	0.153 (1.20)	0.360*** (3.08)	0.141 (1.17)	0.146 (1.15)	0.290 (1.38)	0.521** (2.46)	0.193 (0.96)	−0.174 (−0.80)	0.393* (1.92)	0.654*** (3.35)	0.249 (1.25)	0.050 (0.23)

Note: This table presents the coefficients obtained from DID regressions that examine the impact of data breaches on vendor selection behaviors across different breach scales and time periods surrounding the breaches. We construct a series of time indicators similar to *After Breach* based on the scale of the data breach. For example, *After Breach_1000* is a time-variant indicator that equals 1 for the years that after the first data breach with a scale of less than 1000 records, and 0 otherwise. In Columns (1)–(4), the dependent variable is *Vendor Replacement* measured over different years from 1 year before to 2 years after the breach. In Columns (5)–(8), the dependent variable is *Star Switch* measured over different years from 1 year before to 2 years after the breach. In Columns (9)–(12), the dependent variable is *Single-Sourcing* measured over different years from 1 year before to 2 years after the breach. The sample is a CEM-matched sample, which includes 10,312 observations. All controls are included. Standard errors are clustered by hospital. *t* statistics are in parentheses ($p < .05$; $**p < .01$; $***p < .001$). The highlighted numbers are intended to emphasize the construction of our dependent variable

and in a manner, favorable to information security, such as shifting to star vendors and migrating towards a single-sourcing configuration. However, long-term cooperation with vendors or being part of large health-care systems reduces the tendency to change vendors. In addition, the impact of data breaches on vendor selection behaviors is frequently contingent upon the breach scale and is short-term in nature. These findings have significant theoretical and practical implications, which we will discuss in detail.

7.1 | Theoretical implications

This study makes a significant theoretical contribution. First, it expands on the existing literature on vendor selection, which has traditionally focused on factors such as cost, quality, and delivery capability (Ho et al., 2010; Pal et al., 2013), but has largely neglected the critical aspect of information security. Our study provides empirical evidence for the first time that information security can have a significant impact on a hospital's vendor selection decisions. Moreover, our findings offer valuable insights for future research in supply chain management. It is essential for future studies to consider information security issues and recognize them as a crucial factor in both vendor selection and partner management.

Second, this study enhances the landscape of information security research within the OM domain. While some OM studies have emphasized information security as a critical aspect of supply chain operations (Ulhaq et al., 2016; Williams et al., 2008), they have often neglected to delve into the specific influence of information security on enterprise operations and supply chain decision-making. Furthermore, there remains a scarcity of empirical research on information security within the OM field (Massimino et al., 2018). This study's empirical findings regarding the impacts of information security failures on vendor selection behaviors contribute to bridging these gaps in existing OM research.

Third, this study contributes to the literature on the ABV, which primarily focuses on the factors that influence organizational attention allocation (Joseph & Wilson, 2018; Monteiro et al., 2008) and how it drives organizational actions (Decreton et al., 2017; Vissa et al., 2010). While past research has explored the stimuli of attention within organizations, limited research has been conducted on how negative performance can prompt organizational attention shifting and drive strategic change. By identifying data breaches as an environmental stimulus that prompts organizations to pay more attention to information security and modify their

outsourcing behavior, this study expands upon the existing ABV literature.

Lastly, this study contributes to the literature on multi-sourcing. In academia, the discussion on single-versus multi-sourcing has only just begun. Handley et al. (2022) emphasize the research gap and point out that "surprisingly, limited attention has been paid to the factors that influence multi-sourcing decisions" (p. 3). Our study can help address this research gap by revealing that when organizations encounter information security failures, they frequently opt to forgo the economic advantages of multi-sourcing and transition to a single-sourcing configuration to mitigate security challenges. This finding offers initial evidence that organizations incorporate information security considerations when making decisions between single- versus multi-sourcing strategies. Additionally, our results are intriguing as prior research has indicated that multi-sourcing is a prevalent trend for many organizations (Handley et al., 2022; Levina & Su, 2008). However, our results suggest that security risks may prevent organizations from joining this trend.

7.2 | Managerial implications

In today's digital age, information security has become a major threat to global supply chains. The exponential growth of data has exposed numerous vulnerabilities, resulting in a significant surge in data breaches. The Association for Supply Chain Management (ASCM) has consistently ranked information security among the top 10 supply chain trends for both 2022⁵ and 2023.⁶ In this context, gaining a profound understanding of the role of information security in supply chain relationships and how organizations respond to security failures in their supply chain strategies has become imperative.

Our research provides valuable insights by pinpointing several key trends observed among hospitals in the aftermath of data breaches. These trends encompass a heightened inclination to replace vendors, a shift towards highly-rated vendors, and a transition towards single-sourcing in the wake of data breaches. These findings offer invaluable guidance for vendors to proactively formulate response strategies for customers' security incidents, facilitating them in securing long-term strategic advantages and enhancing their competitiveness.

Furthermore, since vendor replacement typically entails significant costs and operational adjustments, our research underscores the critical need for organizations to prioritize information security in establishing their vendor partnerships. We recommend that organizations undertake thorough risk assessments to precisely gauge the

security risks linked to potential vendors. This approach will enable them to make well-informed decisions and reduce the likelihood of resorting to alternative outsourcing strategies due to inadequate security performance.

Lastly, our study reveals that hospitals with long-term partnerships and those located into extensive healthcare systems are less likely to switch vendors following a data breach. This highlights the importance of nurturing long-lasting client relationships and efficiently serving large healthcare conglomerates. Moreover, our findings offer novel insights into the varying security needs and behavior patterns of clients. These insights enable vendors to develop a customized approach to their client relationships and to effectively adapt and provide necessary support during times of security crisis.

7.3 | Limitations and future research

Like other studies, our research has limitations that create opportunities for future investigation. First, our findings are specific to the healthcare industry and the selection of EMRS vendors. Therefore, future research can delve into how data breaches impact outsourcing strategies in other industries or explore alternative outsourcing strategies in the healthcare sector. Second, while our study provides insights into the US context, organizations' willingness to change vendors after a data breach may be influenced by cultural factors. For example, countries with a high level of uncertainty avoidance tend to have a lower tolerance for uncertainty and may be more risk-averse when it comes to changing vendors after data breaches. In addition, countries with a high degree of individualism prioritize personal autonomy, which may extend to managerial decision-making on vendor selection. Thus, future research could extend our model by incorporating cross-cultural differences. Third, our study has another limitation in that we have only considered changes in strategies within a single organization and have not explored whether data breaches have *spillover* effects on the vendor selection strategies of other organizations. Therefore, future research can further investigate whether the impact of data breaches spreads to the outsourcing strategies of competitors or sister hospitals.

8 | CONCLUSIONS

Our groundbreaking study investigates the impact of data breaches on vendor selection from an ABV perspective. Through a series of rigorous empirical tests, we demonstrate the critical role of information security in

shaping hospitals' vendor selection behaviors. Our research reveals that after a data breach, as decision-makers' attention shifts towards information security, hospitals' original vendor selection strategy tends to be disrupted. Consequently, hospitals tend to change their EMRS vendors; and in the process, they prefer selecting star vendors and migrating towards a single-sourcing configuration. Additionally, we discover that the tendency to replace certain EMRS vendors is mitigated when there is a long-lasting hospital-vendor relationship or when hospitals are part of large hospital systems. Our findings provide a foundation for further investigation of information security issues in the fields of OM and supply chain management.

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ENDNOTES

¹ <https://www.censinet.com/ponemon-institute-research-reveals-majority-of-healthcare-vendors-have-experienced-a-data-breach-exposing-protected-health-information/>.

² KLAS (<https://klasresearch.com>) was established in 1996 by four individuals from the healthcare information technology industry: Kent Gale, Leonard Black, Adam Gale, and Scott Holbrook. Since 2005, KLAS has been presenting the "Best in KLAS Awards," which are specifically conferred upon vendor solutions that lead in market segments encompassing software and services with the most significant operational and clinical impact on healthcare organizations. Vendors' earning of the prestigious title of "Best in KLAS" could signify the recognition of their exceptional efforts to assist healthcare professionals in delivering improved patient care and service.

³ $L1$ distance is an imbalance measure that calculates the absolute difference across all cells in a multivariate histogram. In the context of CEM, the $L1$ norm assesses the dissimilarity in features between individuals before and after matching, providing an evaluation of the matching effectiveness. The $L1$ distance varies between 0 and 1, where a value of 0 represents perfect global balance, while an $L1$ value of 1 indicates perfect imbalance or complete separation of the multidimensional histograms (Iacus et al., 2009; Iacus et al., 2012).

⁴ Weights are commonly employed to deal with non-random or non-probabilistic sample selection. However, in the context of fixed-effects logit model, the estimation relies on within-individual variation, making the use of weights inappropriate. Therefore, we utilize the fixed-effects LPM model in the robustness test (entropy balancing matching).

⁵ <https://www.ascm.org/ascm-insights/10-supply-chain-trends-to-watch-for-in-2022/>.

⁶ <https://www.ascm.org/making-an-impact/research/top-supply-chain-trends-in-2023/>.

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APPENDIX A: EMPIRICAL LITERATURE ON VENDOR SELECTION BEHAVIORS

We review empirical articles on vendor selection behaviors in Table A1.

TABLE A1 Empirical literature on vendor selection behaviors.

Studies	Determinants of vendor selection behaviors	Main insights
Riedl et al. (2013)	Procedural rationality	- The more procedural rationality is used in supplier selection decisions, the less uncertainty there is likely to be. - When decision uncertainty is reduced, supplier decision performance is likely to improve.
Badorf et al. (2019)	Economies of scale	- Client firms' supplier selection decisions are positively influenced by economies of scale, but the impact diminishes as scale increases. - The relationship between economies of scale and supply selection depends on economic, buyer, and relationship characteristics.
Sharland et al. (2003)	Cycle time	Trust and commitment in long-term relationships are determined by a combination of cycle time reduction and consistent high-quality.
Banerjee et al. (2020)	Social media volume and valence	The possibility of supplier selection can be increased by either the volume or valence of social media, but deviations from the reference price weaken the positive association.
Thomas et al. (2021)	Social sustainability	Clients prefer suppliers with desirable levels of social sustainability.
Yan et al. (2020)	Supplier ties	For supplier selection in new product development, managers perceive external innovation ties and other industry customer ties as positive signals for innovation novelty, but competitor ties as negative signals. However, all three types of ties are perceived as negative signals for information protection.
Ta et al. (2018)	Trustworthiness	- Changes in trustfulness can influence relationship continuity after contract breaches, but this influence is not linear. - The timing or security of contract breaches moderates the relationship between trustfulness and relationship continuity.
Wang et al. (2022)	Responsibility attribution of supplier-induced disruptions	- Distributive justice weakens the relationship between disruptions triggered by suppliers' internal incidents and clients' switching intentions. - Procedural and interactional justice weakens the relationship between disruptions triggered by suppliers' external incidents and clients' switching intentions.
Polyviou et al. (2018)	Clients' post-disruption emotion	- When a supply disruption occurs, a recovery lead is likely to recommend against sourcing from the incumbent critical-component supplier if they believe that the supplier, rather than nature, had control over the disruption. - The recovery lead's level of negative emotion resulting from the supply disruption may partially explain the impact of their recommendation.
Whitten and Leidner (2006)	Relationship quality and switching cost	- The decision to back-source application outsourcing is determined by a combination of product quality, service quality, relationship quality, and switching costs. - When considering a vendor switch, relationship quality and switching costs are the two main factors that influence the decision.

APPENDIX B: EMR MODULES

EMRS is an application environment comprising multiple distinct yet interconnected modules. As new components are developed, and old ones may be phased out or merged into others, the composition of EMR modules could evolve over time. Despite the absence of a universal consensus on EMRS composition, we have observed that a substantial body of business academic research consistently places emphasis on the following five modules: Clinical Data Repository (CDR), Clinical Decision Support System (CDSS), Computerized Provider Order Entry (CPOE), Order Entry (OE), and Physician Documentation (PD) (e.g., Angst et al., 2017; Ganju et al., 2022; Greenwood et al., 2019; Hydari et al., 2019; Sharma et al., 2016). These modules are extensively utilized and encompass the fundamental functionalities required for daily care and medical management in healthcare institutions. In our study, we have adhered to this academic convention and considered these five modules. Table B1 provides the respective definitions of these five modules, sourced from a subscription-based proprietary database (HIMSS Analytics, 2015).

TABLE B1 EMR module.

EMR module	Definition
CDR	A centralized database that allows healthcare organizations to collect, store, access, and report on clinical, administrative, and financial information collected from various applications within or across the organization.
CDSS	An application that uses pre-established rules and guidelines, that can be created and edited by the healthcare organization, and integrates clinical data from several sources to generate alerts and treatment suggestions.
CPOE	An order entry application specifically designed to assist clinical practitioners in creating and managing medical orders for patient services or medications.
OE	An application that allows for entry of orders from multiple sites including nursing stations, selected ancillary departments, and other service areas; allows viewing of single and composite results for each patient order.
PD	A software that documents physician notes that describe the care or service to a patient.

APPENDIX C: EXAMPLD BREACH DESCRIPTION AND TYPE

TABLE C1 Exempld breach description and type.

Source	Organization name	Breach year	Description of exempld breach (from PRC or ITRC)	Breach type
PRC	Boston Scientific	2014	It has been reported that Boston Scientific, a medical device maker's computer network, has been hacked sometime in the first half of 2013. It is not clear what type of information the hackers were targeting. Federal laws meant to safeguard medical information require companies to disclose any breach involving patient information, so far Boston Scientific has not made these disclosures. Denise Kaigler, a Senior Vice President of Corporate Affairs with Boston Scientific, stated "like many companies, Boston Scientific experiences attempts to penetrate our networks and systems, and we take such attempts seriously. We have a dedicated team to detect and mitigate attacks when they occur, as well as to implement solutions to prevent future attacks." Ms. Kaigler would not comment on the specifics of any attack but described the media reporting as "inaccurate." The attacks point to Chinese hackers, and the medical device company was not aware of the intrusions until federal authorities contacted them. They have now formed a task force to investigate the breach.	External Attacks
ITRC & PRC	Behavioral Health Network	2013	You'd like to think your medical records are safe but the 22News I-Team discovered today a number of personal documents left in a dumpster for anyone to see.	Internal Breach
ITRC & PRC	Boca Raton Regional Hospital	2013	Eight people were charged for participating in an identity theft ring. One of the members was employed as a scheduler at Boca Raton Regional Hospital. She passed along patient information in exchange for payments. One member allegedly filed 57 fraudulent tax returns with the stolen information in attempt to get \$306,720 in refunds. Another member is accused of filing 75 fraudulent returns for \$750,469 in refunds. UPDATE (07/30/2013): The dishonest employee was convicted of unauthorized disclosure of medical records, using stolen Social Security numbers to file fraudulent tax claims, and conspiracy to commit false claims. She was sentenced to 18 months in prison and ordered to pay \$15,795 in restitution to the IRS. A co-defendant was sentenced to 40 months for her role in filing fraudulent tax returns and stealing patient information.	Internal Breach
ITRC	Child & Family Psychological Services, Inc.	2013	Hacking/IT incident —network server	External Attacks

Abbreviations: ITRC, Identity Theft Resource Center; PRC, Privacy Rights Clearinghouse.

APPENDIX D: VARIABLE DESCRIPTIONS

Table D1 provides the definitions and data sources of the variables included in the baseline analysis.

TABLE D1 Variable descriptions.

Variable	Description	Data source
<i>Replacement</i>	A dummy variable that equals 1 if a hospital replaces its vendor in any of the EMRS modules (i.e., CDR, CDSS, CPOE, OE, or PD) in a certain year and 0 otherwise	HIMSS
<i>Star Switch</i>	A dummy variable that equals 1 if a hospital switches to a star vendor in any of the EMRS modules (i.e., CDR, CDSS, CPOE, OE, or PD) in a certain year and 0 otherwise	HIMSS
<i>Single-Sourcing</i>	A dummy variable that equals 1 if a hospital's distance from the single-sourcing configuration is reduced compared with the one in the last year and 0 otherwise	HIMSS
<i>After Breach</i>	A time-variant indicator that equals 1 if an observation is in the year after the first data breach during our sample period and 0 otherwise	PRC, ITRC
<i>Breach</i>	A treatment group indicator that equals 1 if a hospital is in the treatment group and 0 if it is in the control group	PRC, ITRC
<i>Duration</i>	Logarithm of the maximum number of years that a hospital has contracted with the current vendor for a specific module among the EMRS modules (i.e., CDR, CDSS, CPOE, OE, or PD)	HIMSS
<i>System Size</i>	Logarithm of the total number of affiliated hospitals in a hospital's parent healthcare group	HIMSS
<i>Hospital Size</i>	Logarithm of a hospital's licensed bed number	HIMSS
<i>Hospital Type</i>	A categorical variable that represents the type of a hospital with the following mappings: 1 for "Academic," 2 for "Acute Psychiatric," 3 for "Acute Rehabilitation," 4 for "Cardiology," 5 for "Critical Access," 6 for "Eye, Ear, Nose & Throat," 7 for "General Medical," 8 for "General Medical & Surgical," 9 for "Long Term Acute," 10 for "Microhospital," 11 for "Neurology," 12 for "Oncology," 13 for "Orthopedic," 14 for "Other Specialty," 15 for "Pediatric," 16 for "Pediatric, Women's Health," 17 for "Psychiatric," and 18 for "Women's Health"	HIMSS
<i>Revenue</i>	A hospital's net operating revenue (in units of 100 million)	HIMSS
<i>Profit Margin</i>	The percentage of a hospital's net profit out of its total revenue	
<i>Digitalization Degree</i>	A categorical variable that indicates the percent range of a hospital's current medical record that is electronic: 0 if the percent is below 25%, 1 if the percent is between 26% and 50%, 2 if the percent range is between 51% and 75%, and 3 if the percent is between 76% and 100%	HIMSS
<i>IS Expense Intensity</i>	The percentage of a hospital's Information System (IS) department operating expense out of its total operating expense	HIMSS

APPENDIX E: PLACEBO TEST

In this appendix, we explain how we carry out the placebo test. Our regression sample initially contains 448 breached hospitals. First, we randomly select another 448 non-breached hospitals to create a “fake” treatment group, and randomly select another 448 non-breached hospitals as a control group. This process allows us to construct a “fake” treatment group indicator ($Breach_{fake}$) that equals 1 if the hospital is in the “fake” treatment group and 0 if in the control group. Second, we assign the data breach years of the 448 true breached hospitals to each “fake” breached hospital, creating a “fake” time-

variant indicator ($After\ Breach_{fake}$) that equals 1 if the time period is the post-breach period and 0 if a pre-breach one. Third, we construct a fake variable of interest, $Breach_{fake} \times After\ Breach_{fake}$. Randomization is used to ensure that the newly constructed regressor does not have an effect on the dependent variables, namely *Replacement*, *Star Switch*, and *Single-Sourcing*. Hence, if there are no significant omitted variables, then the t -value for $Breach_{fake} \times After\ Breach_{fake}$ should not deviate significantly from 0. At last, to avoid contamination by rare events, we repeat this random data-generating process 500 times and plot the distribution of 500 t -values of the “fake” double-difference term in Figure 3 of the main text.

APPENDIX F: ENTROPY BALANCING

During the entropy balancing matching process, we include all continuous control variables (i.e., *Hospital Size*, *Revenue*, *Profit Margin*, *IS Expense Intensity*, and *System Size*) as the covariates. Additionally, we employ the balancing method to ensure that the means, variances, and skewness of these variables are nearly identical between the balanced control group and the treatment group.

Table F1 summarizes the results of entropy balancing. Columns (1) and (2) show the means and skewness of the covariates in the treatment group, whereas Columns

(3) and (4) provide the corresponding information for the control group. In the pre-balancing sample, these statistics show substantial differences. However, in the post-balancing sample, the balanced control group's statistics are almost identical to those of the treatment group. We also regress the treatment variable on these variables and presented the results for both the unbalanced and balanced samples in Column (5) to examine whether these variables could predict the allocation of the treatment. The results show that the coefficients of these variables became smaller, lost statistical significance, and the *pseudo R*² decreased from 0.119 to 0.002. This indicates that after balancing, these variables are no longer able to predict the treatment allocation.

TABLE F1 Balancing test results of entropy balancing method.

	Treatment		Control		Logit
	Mean	Skewness	Mean	Skewness	
	(1)	(2)	(3)	(4)	(5)
Before balancing					
<i>Hospital Size</i>	5.336	−0.765	4.388	0.077	−0.062 (−1.96)
<i>Revenue</i>	459.300	2.170	135.800	5.327	0.775 (29.28)
<i>Profit Margin</i>	−0.328	−5.317	−0.131	−8.063	−0.218 (−17.52)
<i>IS Expense Intensity</i>	0.005	3.637	0.002	96.580	1.580 (2.44)
<i>Healthcare System Size</i>	1.966	0.860	2.076	0.718	−0.184 (−15.42)
<i>Pseudo R</i> ²					0.119
After balancing					
<i>Hospital Size</i>	5.336	−0.765	5.336	−0.765	0.058 (1.26)
<i>Revenue</i>	459.300	2.170	459.300	2.170	0.035 (0.86)
<i>Profit Margin</i>	−0.328	−5.317	−0.328	−5.317	0.001 (0.09)
<i>IS Expense Intensity</i>	0.005	3.637	0.005	3.850	0.036 (0.02)
<i>Healthcare System Size</i>	1.966	0.860	1.966	0.860	−0.034 (−2.04)
<i>Pseudo R</i> ²					0.002

Note: This table presents the results of entropy balancing. Columns (1) and (2) present the means and skewness of variables in the treatment groups, while Columns (3) and (4) provide the same statistics for the control group. Column (5) reports the logit regression results of regressing the treatment dummy on the variables before and after balancing. *t* statistics are in parentheses.